
CS614 - Midsem Project Review

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Project Overview

Iterative pre-copy live migration transfers memory pages from a running process to a destination host while the process continues to execute, repeating the transfer for pages that were dirtied during the previous round until the remaining working set is small enough to complete migration with a brief final pause. The efficiency of this scheme depends entirely on the ability to identify, at the end of each round, *exactly* which pages were written. For CPU memory, the Linux kernel provides this capability through soft-dirty bits embedded in page-table entries, readable via `/proc/PID/pagemap`. No analogous mechanism exists for GPU-managed memory. NVIDIA's Unified Virtual Memory (UVM) subsystem, which underpins `cudaMallocManaged` and handles demand-paging between host and device, exposes no interface for querying which pages a process has modified. Consequently, any migration or checkpointing system must conservatively copy the *entire* GPU memory footprint on every iteration, rendering iterative pre-copy migration of large GPU workloads impractical.

This project addresses that gap by implementing **per-process dirty page tracking** directly within the open-source NVIDIA UVM kernel module. UVM already intercepts every GPU page fault to manage demand-paging; we augment the write-fault servicing path to record each written page in an in-kernel data structure, without introducing any additional hardware overhead. A `procfs`-based interface allows a privileged migration daemon to start and stop tracking epochs and to retrieve the complete set of pages written by a given process, with **page-level granularity** and a per-page **nanosecond-resolution timestamp** suitable for ordering events across migration rounds.

Group Details

Following are the details of the group members:

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Milestones

Milestone 1 - Midsem Review

(4 weeks)

1. Literature Review, Technical Familiarization and other Preliminaries

(1 week)

- Review existing technical documentation and architectural specifications regarding UVM.
- Conduct a code review of the UVM driver and perform preliminary API experimentation.

2. Implementation of Dirty Access Tracking in UVM Driver

(2 weeks)

- Add instrumentation and logging to observe runtime behavior, including tracking page faults, permission transitions, migration events, and residency changes for debugging and analysis.

- (b) Implement the core dirty access tracking mechanism, including data structures for tracking dirty pages, and the logic to update these structures on write faults.
- (c) Develop a `procfs`-based interface to allow user-space tools to query the set of dirty pages for a given process, including timestamps for each page.

3. Initial Performance Evaluation (1 week)

- (a) Evaluate the performance of the modified driver on small workloads, measuring metrics such as page fault latency, migration throughput, and overall application runtime.
- (b) Identify and address any significant performance bottlenecks or correctness issues in the implementation.

Milestone 2 - Extensions, Testing and Optimisations (4 weeks)

1. Extension of Functionality (2 week)

- (a) Extend the tracking mechanism to support additional features such as multiple concurrent processes, concurrent tracking on CPU and GPU side, and removing write permission for the tracked pages at init, instead of invalidating them to reduce overhead due to read-only GPU pages
- (b) Extend the `procfs` interface to expose richer statistics, including per-page access frequency counters and dirty page age, to enable more informed migration and eviction decisions by user-space tools.

2. Testing and Analysis of Tracking Mechanism (1 week)

- (a) Test critical issues pertaining to concurrency and performance that arise under multi-process workloads, including correctness of dirty page tracking under concurrent write faults and race conditions in shared data structures.
- (b) Quantify overhead introduced by dirty tracking, measuring additional latency per write fault due to metadata updates, and memory footprint of tracking data structures such as counters, hash tables, and bitmaps.
- (c) Profile driver performance across diverse workloads with and without dirty tracking enabled, for metrics such as page fault latency, migration throughput, and overall application runtime.

3. Optimisation of Tracking Mechanism (1 week)

- (a) Based on the analysis, identify and implement optimizations to reduce overhead, such as batching updates to tracking data structures, using more efficient data structures (e.g., bitmaps instead of hash tables), and optimizing critical paths in the tracking logic.

Milestone 3 - Final Deliverables and Integration (2 weeks)

1. Prepare comprehensive documentation for the tool, including usage guidelines and performance considerations.
2. Produce formal benchmarking results quantifying the tracker's performance impact across various scenarios.
3. Implement user-facing configurability for the tracker via a command-line interface (CLI).

Source Code

Development Environment

Development and testing are conducted on a dedicated virtual machine (`dirtyvm`) running:

- Linux kernel version 6.8 (custom build)
- Modified NVIDIA UVM driver version 580.65.06 (loaded as a kernel module)

Repository

The full source code is hosted at:

<https://github.com/aleph-5/open-gpu-kernel-modules>

This is a fork of NVIDIA's open-source GPU kernel modules, with our modifications applied on top.

Modified Driver Files

All driver modifications reside under `kernel-open/nvidia-uvmm/`. The files containing functionality added or significantly modified by us are:

1. `uvmm_common.h / uvmm_common.c` - The primary location of our implementation. Defines `struct dirty_page_info` (page number, timestamp, PID) and `struct uvmm_dirty_page_table` (an `xarray`-based per-process table). Implements the core API: table initialisation and teardown, fault recording, and page lookup. Also declares `uvmm_dirty_procfs_init` for the `procfs` interface.
2. `uvmm_va_block.c` - Hooked to invoke the dirty tracking record function when a write fault is serviced on a VA block.
3. `uvmm_gpu_replayable_faults.c` - Modified to record dirty page events when GPU write faults are replayed.
4. `uvmm_va_space.c / uvmm_va_space.h` - Extended to register the dirty-page GPU-unmap invalidation callback (`uvmm_va_space_dirty_init`) and manage per-process tracking table lifetime alongside VA space creation and destruction.
5. `uvmm.c` - Top-level integration point; initialises the `procfs` dirty tracking interface at module load.

Tests

User-space tests are divided categorized into correctness and performance tests and located in the `correctness_tests/` and `performance_tests/` directories respectively

The correctness tests are run everytime the driver is modified and reloaded using `run_regression.sh`

Current Progress

Implementation of Dirty Access Tracking in UVM Driver

Workflow A tracking epoch begins when a privileged user-space tool writes to `dirty_pids_start_track`. This initialises a fresh per-process table and immediately invalidates all current GPU page-table entries by unmapping them, so that every subsequent GPU access generates a fresh page fault. Each write fault is intercepted in the driver's fault-servicing path, which records the faulting page in the table along with a nanosecond-resolution timestamp before re-establishing the mapping and replaying the faulting instruction. At any point during the epoch the tool can read `dirty_pages` (provided that the PID has been set using `dirty_pid_to_query`) to obtain the accumulated set of written pages; reads are non-destructive, so the table is not consumed by the query. The epoch ends when the tool writes to `dirty_pids_stop_track`, which frees all recorded state. Re-starting tracking destroys any existing table before creating a fresh one, cleanly beginning a new epoch.

Data Structures The tracking state is organised as a two-level structure defined in `uvmm_common.h`.

1. `struct dirty_page_info` - a per-page record holding the page number, the `ktime_get_ns()` timestamp of the first observed write fault, and the PID of the faulting process.
2. `struct uvmm_dirty_page_table` - a per-process container holding an `xarray` that maps page numbers to `dirty_page_info` pointers.
3. A global `xarray (pid_to_page_table)` maps each tracked process's `tgid` to its `uvmm_dirty_page_table`.

This two-level design provides natural per-process isolation. Entries are recorded under a *first-write-wins* policy: an existing entry is never overwritten, so the stored timestamp always reflects the earliest observed write to that page within the current epoch. The faults may be serviced by different CPU-side driver threads, thus the timestamp may not be strictly monotonic, depending upon the driver-side service thread recording the same.

APIs Exposed User-space interacts with the tracker exclusively through five `procfs` entries created under `/proc/driver/nvidia-uvmm/`:

1. `dirty_pids_start_track` (write) - initialises (or re-initialises) the tracking table for `current->tgid` and triggers GPU-mapping invalidation to start a new epoch.
2. `dirty_pids_stop_track` (write) - destroys the table for `current->tgid`, freeing all recorded entries.
3. `dirty_pid_to_query` (write) - sets the PID whose table is consulted on subsequent `dirty_pages` reads, decoupling the querying process from the tracked process.

4. `dirty_pages` (read) - iterates the selected process's table and emits one line per recorded page in the format `0xaddr timestamp_ns pid`. Reads are *non-destructive*: entries remain in the table and can be queried repeatedly within the same epoch.
5. `dirty_range` (write) - accepts a pair of hexadecimal addresses `0xstart 0xend` (end exclusive) to restrict subsequent `dirty_pages` reads to a specific virtual address window. The range is protected by a spinlock to allow concurrent updates safely. An inverted or zero-length range produces an empty result rather than undefined behaviour.

Internal Functions and Implementation Sites The implementation is spread across several driver files, each responsible for a distinct concern:

1. `uvm_common.c` - the primary implementation file. Contains `uvm_dirty_page_table_init` / `uvm_dirty_page_table_destroy` (table lifecycle), `uvm_dirty_page_table_record` (fault recording, using `GFP_ATOMIC` as it executes in a non-sleepable context), `uvm_dirty_page_table_lookup`, and the full `procfs` handler implementations including `uvm_dirty_procfs_init`.
2. `uvm_gpu_replayable_faults.c` - the write-fault hook. After standard fault servicing, checks `uvm_dirty_tracking_active_for_pid(va_block->creator_pid)` and calls `uvm_dirty_page_table_record` for `UVM_FAULT_ACCESS_TYPE_WRITE` faults.
3. `uvm_va_space.c` - implements `uvm_dirty_invalidate_all_gpu_mappings`, which walks all live VA spaces and blocks to unmap GPU PTEs at epoch start. Exposes `uvm_va_space_dirty_init` to register this function via the `uvm_dirty_invalidate_fn` function pointer at module load.
4. `uvm.c` - top-level integration; calls `uvm_dirty_procfs_init` and `uvm_va_space_dirty_init` during module initialisation.

Correctness Tests

At this stage, evaluation is driven by the CUDA tests under `correctness_tests/`. The suites are intended to cover four broad questions:

1. **Generic sanity checks:** The `generic` suite is the basic end-to-end test for dirty-page tracking: it GPU writes are recorded, read-only accesses are not, and that reset, disable, PID reporting, and range filtering behave as expected.
2. **Table lifecycle:** The `table_lifecycle` suite checks whether starting, stopping, and restarting tracking creates a clean epoch each time without stale entries leaking across resets.
3. **Ordering and concurrency:** The `ordering` suite checks whether timestamps and page membership stay reasonable when writes arrive across epochs, across repeated queries, and under high-concurrency GPU execution.
4. **Address-range filtering:** The `address_range_filtering` suite checks whether `dirty_range` correctly limits query results and whether query-time range filtering behaves correctly across valid and invalid ranges.

Observations The tests were rerun as root, and the current implementation passed every test that is still present in the repository. The aggregate result was **30/30 passed: table_lifecycle 5/5, ordering 8/8, address_range_filtering 5/5**, and the `generic` suite 12/12. The per-test runners for the first four suites report runtimes between 1.94s and 2.07s.

Suite	Test	Purpose	Result	Time
Generic	T01_writes_recorded	a GPU write marks all managed pages dirty.	PASS	–
Generic	T02_reads_not_recorded	read-only GPU accesses do not create dirty entries.	PASS	–
Generic	T03_reset_clears_table	resetting tracking removes pre-reset entries.	PASS	–
Generic	T04_mixed_read_write	read-only pages stay clean while written pages are tracked.	PASS	–
Generic	T05_migration_write_retracked	pages are tracked again after CPU migration and a fresh GPU write.	PASS	–
Generic	T06_tracking_disabled	<code>procfs</code> reports tracking inactive when tracking is disabled.	PASS	–
Generic	T07_pid_recorded	each recorded dirty entry carries the creator PID.	PASS	–

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Suite	Test	Purpose	Result	Time
Generic	T08_same_page_single_entry	repeated writes to the same page do not create duplicate entries.	PASS	–
Generic	T09_range_inside	a range exactly covering the allocation returns all pages.	PASS	–
Generic	T10_range_outside	a disjoint range returns no dirty pages.	PASS	–
Generic	T11_range_partial_coverage	only the in-range half of an allocation is returned.	PASS	–
Generic	T12_range_boundary_pages	Checks inclusive/exclusive boundary handling for the configured range.	PASS	–
Table lifecycle	tc01_basic_lifecycle	Verifies that a normal start-query-stop tracking cycle works.	PASS	1.98 s
Table lifecycle	tc02_reinit_clears_entries	restarting tracking clears the previous epoch’s entries.	PASS	1.95 s
Table lifecycle	tc03_stop_start_empty_table	stop followed by restart begins from an empty table.	PASS	1.95 s
Table lifecycle	tc04_multi_reinit_stress	Repeats restart cycles to ensure only the current epoch remains visible.	PASS	1.98 s
Table lifecycle	tc05_two_allocs_one_table	two allocations within one process share the same tracking table correctly.	PASS	1.97 s
Ordering	tc01_write_before_start	Confirms writes issued before tracking starts are not reported later.	PASS	1.97 s
Ordering	tc02_ts_monotonic	Checks whether recorded timestamps remain sensible across per-page writes.	PASS	1.97 s
Ordering	tc03_query_nondestructive	Verifies that reading dirty_pages does not consume existing entries.	PASS	1.95 s
Ordering	tc04_empty_table_active	Checks query behaviour when tracking is active but no page is dirty yet.	PASS	1.96 s
Ordering	tc05_epoch_isolation	Verifies that pages from one epoch do not leak into the next.	PASS	1.97 s
Ordering	tc06_concurrent_streams	Stresses concurrent page recording with multiple CUDA streams writing in parallel.	PASS	1.99 s
Ordering	tc07_sentinel_flood	an early sentinel write survives a later flood of writes.	PASS	1.98 s
Ordering	tc08_flood_between_queries	two heavy write phases remain visible across intermediate queries.	PASS	1.99 s
Address range	tc01_two_alloc_range_isolation	Verifies that the configured range selects one allocation and excludes another.	PASS	1.98 s
Address range	tc02_range_sequential_reads_threaded	Checks stable filtered output while different subranges are read in sequence.	PASS	1.94 s
Address range	tc03_subpage_start_rounding	a sub-page range start rounds to the expected page boundary.	PASS	1.95 s
Address range	tc04_inverted_and_zero_range	Verifies that invalid or empty ranges do not return spurious entries.	PASS	1.99 s
Address range	tc05_filter_before_start_track	Checks whether a range set before tracking starts is applied correctly.	PASS	1.99 s

Current interpretation These results give initial evidence that the current implementation is already stable across epoch lifecycle management, non-destructive querying, range filtering and PID tagging checks. A fuller evaluation is still needed for larger dirty sets and explicit overhead measurements, but the present test battery now provides a credible behavioural baseline for the next phase of performance work.

Performance Tests

At this stage, evaluation is driven by the CUDA tests under `tests/performance_testing`. These tests cover the following categories

1. **Stress Testing:** The `tc01_stress_testing` suite verifies that no dirty pages are missed under high write pressure. A single `cudaMallocManaged` allocation of N pages (default $N = 1000$, configurable via command-line argument) is first warmed up with a GPU write kernel so that all pages are resident on

the device before tracking begins. Tracking is then started, which invalidates all GPU page-table entries and forces a fresh write-fault for each page on the subsequent kernel launch. After the kernel completes, `dirty_pages` is queried and the recorded count is compared against N ; the test passes if and only if every page has been recorded. The suite also prints baseline and tracked kernel runtimes together with the per-run invalidation latency.

2. **Performance Overhead:** This tracks the overhead that our solution introduces across the following workload types

- **READ:** We only do read accesses
- **WRITE:** We only do write accesses
- **MIXED:** Half of the access are reads and the other half are writes

This test runs a simple workload across different number of pages and different number of iterations of repeating these access. This workload is then run with dirty tracking active and then the 2 times are compared. The results can be found in `results.csv` To simulate environments when the tracking is activated when a program is already running pages are prefetched before starting the benchmarking

Observations The tests were run as root and although the current implementation correctly tracked all pages under high write loads, the performance overhead was extremely high reaching as high as 1000% under *Write only* loads.

Suite	Test	Purpose	Result
Performance	<code>tc01_stress_testing</code>	under heavy loads no pages are missed	PASS
Performance	<code>tc02_performance_overhead</code>	check the performance penalty imposed by our dirty tracking approach	Check <code>results.csv</code>

Performance Analysis

Stress Testing

The figures below are derived from the `WRITE` rows of `results.csv` after running `tc01_stress_testing` (averaged across iteration counts of 10, 50, and 200).

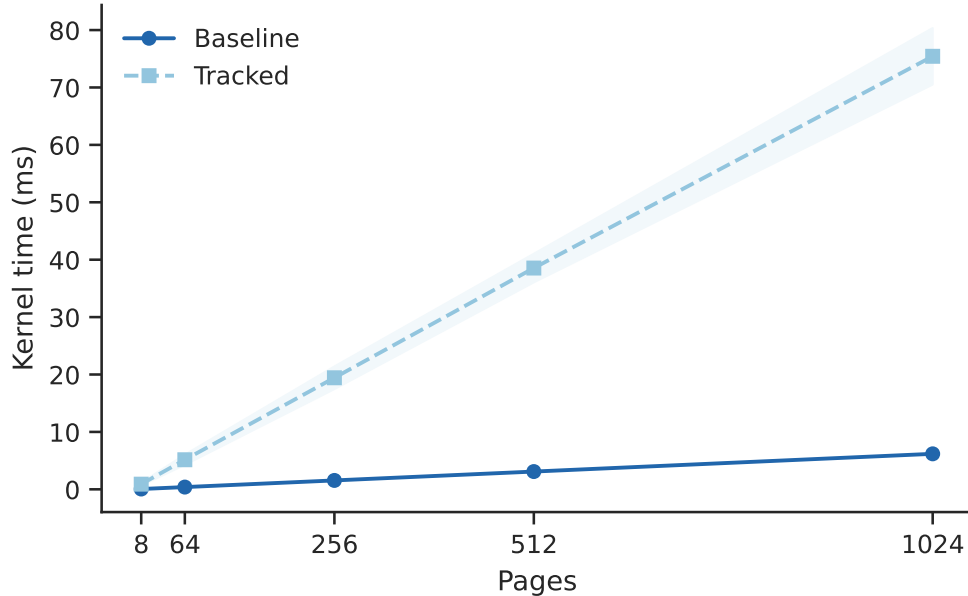


Figure 1: Kernel execution time (ms) with and without dirty-page tracking as a function of allocation size. Both curves scale linearly with page count; the tracked curve is roughly 12 $\times$ higher, reflecting the per-page fault-servicing cost added by the UVM write-fault hook. Shaded bands show $\pm 1\sigma$ across the three iteration counts.

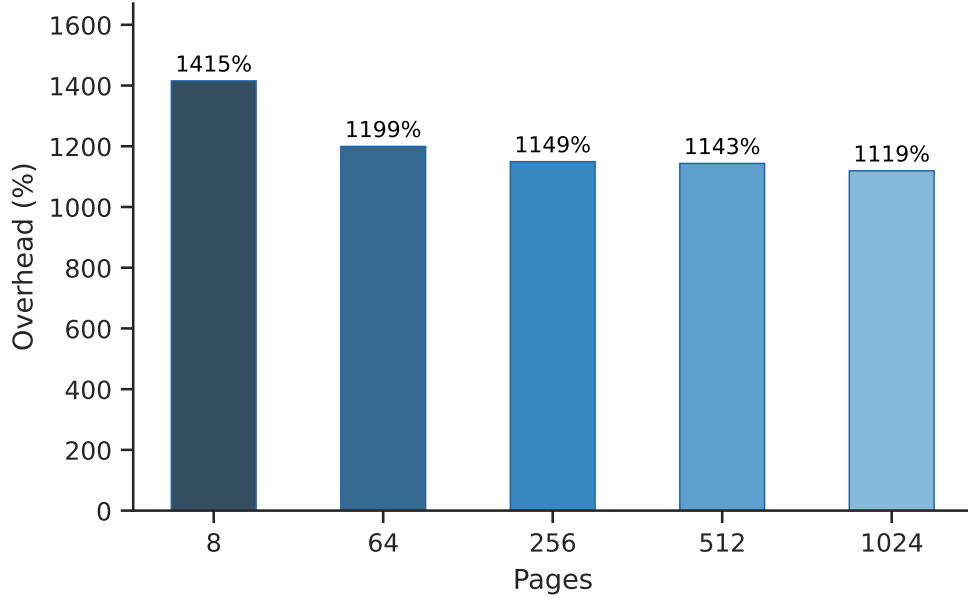


Figure 2: Wall-time overhead introduced by dirty-page tracking for a pure-write workload. The overhead is highest ($\approx 1725\%$) at small allocations (8 pages), where the fixed cost of invalidation and fault-handling dominates, and converges to roughly 1090–1160% for larger allocations where per-page costs dominate. This confirms that the overhead scales proportionally with the number of tracked pages rather than being dominated by a fixed startup cost.

Performance Overhead (tc02)

`tc02_performance_overhead` benchmarks three workload types—READ, WRITE, and MIXED—across five page counts with tracking on and off, averaging results over 10, 50, and 200 iterations.

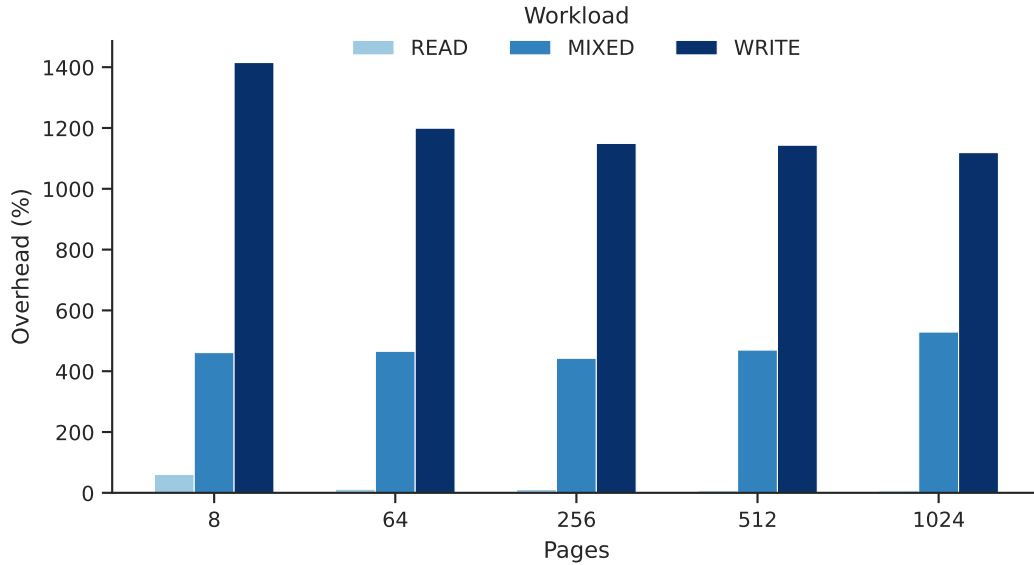


Figure 3: Wall-time overhead (%) by workload type and page count. READ incurs only ~ 6 –95% overhead (no write faults), MIXED ~ 415 –497%, and WRITE ~ 1090 –1725%, confirming that overhead is proportional to the number of write-faulting pages

Duplicate Fault Skip Cost

When a page already recorded in the dirty table faults again, the driver takes an early-return path (first-write-wins) instead of re-inserting the entry. This microbenchmark isolates the cost of that skip path by running the same write kernel twice within one epoch across 100 iterations.

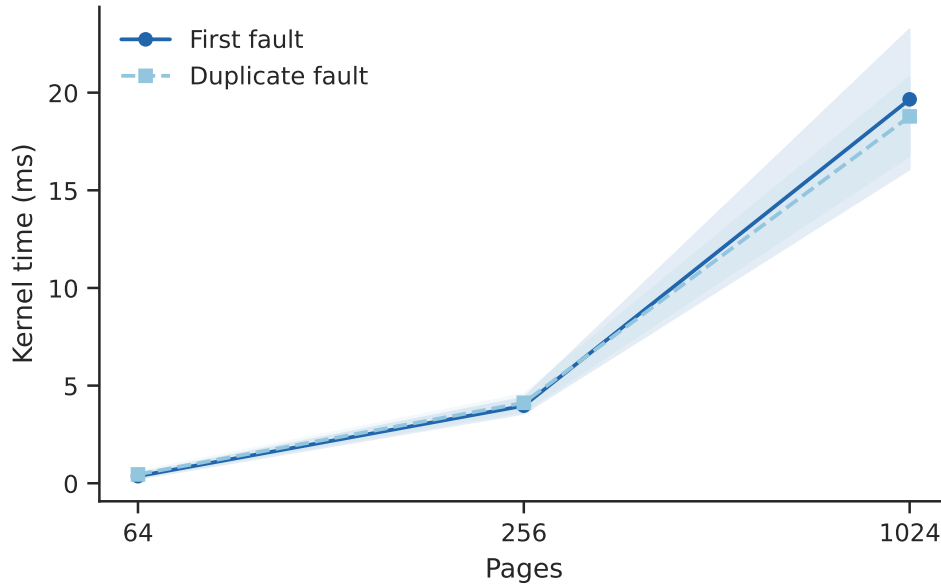


Figure 4: Mean kernel time for first and duplicate faults vs. page count ($\pm 1\sigma$, log-scale x-axis). Both curves track each other closely across all page counts

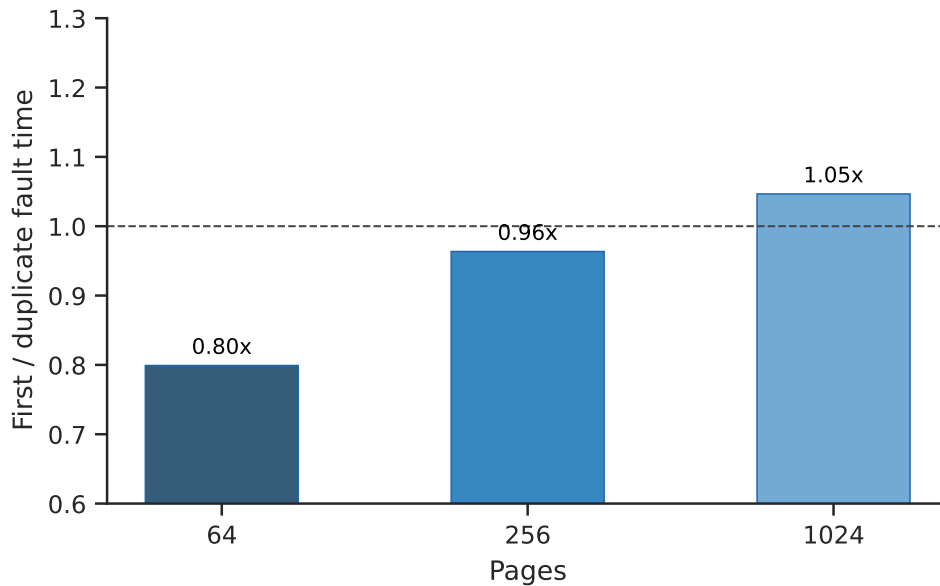


Figure 5: Ratio of first to duplicate fault time. Values near 1.0 (dashed line) indicate that the xarray lookup in the skip path costs nearly as much as a full first-fault recording, suggesting the bottleneck is the fault-servicing path itself rather than the table insertion

Record Contention Cost

This microbenchmark runs multiple CUDA streams concurrently under two conditions: *disjoint*, where each stream writes to a separate page range, and *hot set*, where all streams write to the same pages. Stream counts of 1, 2, 4, and 8 are tested across four page counts over 100 iterations each.

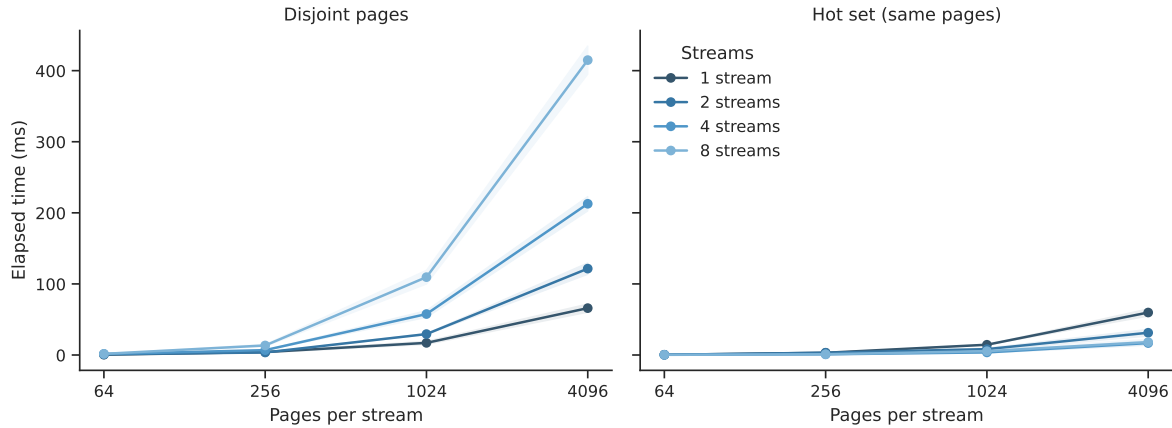


Figure 6: Elapsed time (ms) vs. pages per stream for disjoint and hot-set conditions, by stream count. Hot-set times are consistently lower because concurrent streams hitting the same pages trigger the first-write-wins skip path, reducing total recording work

Table Lifecycle Cost

This microbenchmark times the three procs operations that manage the tracking table: **start_track** (init, allocates an empty xarray and invalidates N GPU PTEs), **stop_track** (destroy, walks and frees all N recorded entries), and a second **start_track** while the table is populated (reinit which is destroy + invalidate N write PTEs + M read PTEs).

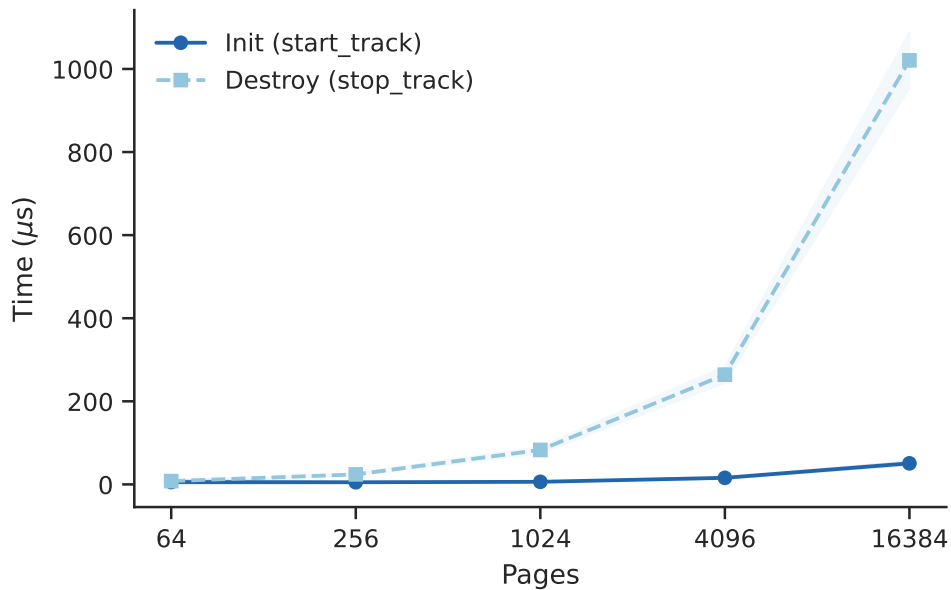


Figure 7: Mean init and destroy time (μs) vs. page count ($\pm 1 \sigma$, log-scale x-axis). Init is nearly flat at 5–51 μs because the xarray allocation is $O(1)$ and PTE invalidation is fast. Destroy scales linearly from 8 μs to ≈ 1 ms at 16384 pages, as it must free one heap allocation per recorded entry.

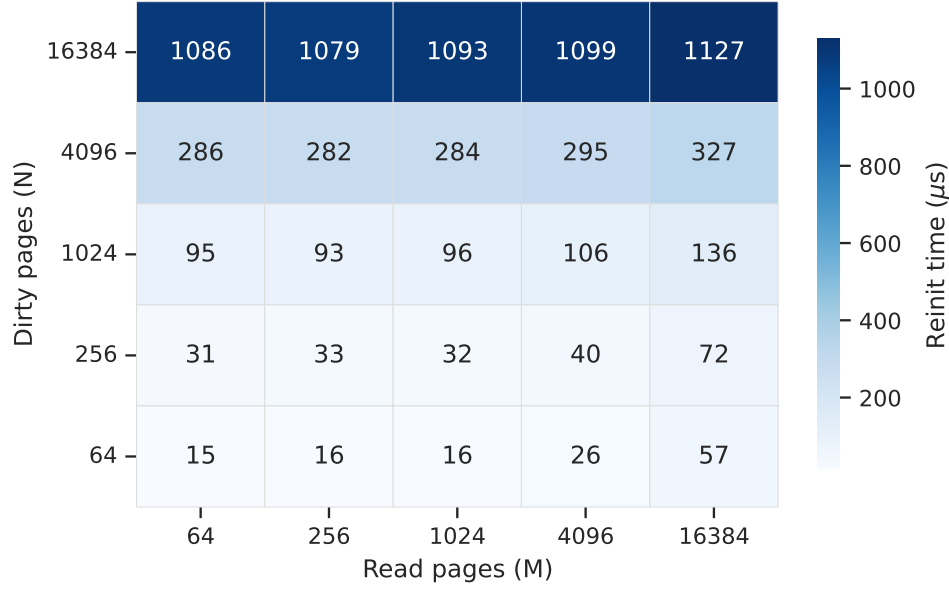


Figure 8: Mean reinit time (μs) as a function of dirty pages (N , rows) and read pages (M , columns). Reinit combines destroy cost (dominant, grows with N) and PTE invalidation (grows with $N + M$), so the cost rises steeply along the N axis and more gently along the M axis.

Single Fault Record Cost

This microbenchmark isolates the per-fault recording cost with a single-thread kernel that writes 4096 pages serially, run 100 times with and without tracking active.

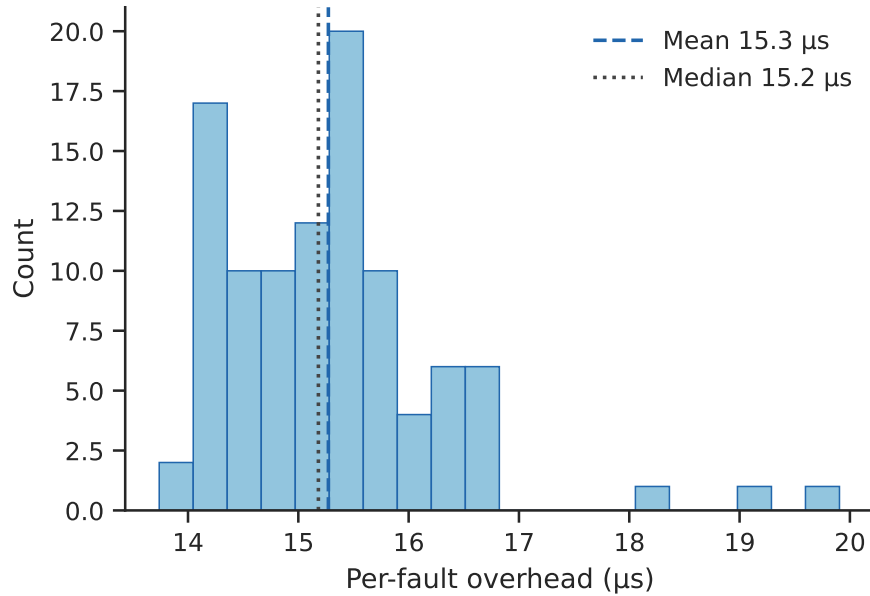


Figure 9: Per-fault overhead derived as $(\text{tracked} - \text{baseline}) \times 10^3 / 4096$. The distribution is tightly concentrated around a mean of $\approx 15.3 \mu s$ per fault (dashed), indicating a stable and predictable per-page recording cost

Next Steps and Planned Progress

Multi-process isolation and range-based dirty tracking (the `dirty_range` procfs interface) are fully implemented. The test suite covers table lifecycle, ordering guarantees, address range filtering, and fault-to-visibility latency. The remaining work is as follows.

Extensions Still Pending

1. **CPU-side Dirty Tracking.** The current implementation intercepts GPU write faults only. Extending tracking to CPU-side writes requires integration with the `mmu_notifier` invalidation path so that CPU dirty pages are recorded in the same per-process table and exposed through the existing procfs interface.
2. **Per-page Access Frequency Counters.** The tracker currently operates under a first-write-wins policy: each page is recorded once with its earliest observed write timestamp. Richer statistics, such as per-page write counts and dirty page age, are needed to support informed migration and eviction decisions by user-space tools. This requires extending `struct dirty_page_info` with a counter field and updating the fault hook to increment it atomically.

Testing and Performance Analysis

1. **Overhead Quantification.** The per-fault overhead introduced by the tracking path (xarray lookup and conditional insert under `GFP_ATOMIC`) has not been formally benchmarked. Write fault latency will be measured with and without tracking enabled across a range of workload sizes to bound the cost.
2. **Memory Footprint Profiling.** Under workloads with large dirty sets, the two-level xarray structure may incur non-trivial memory overhead. The footprint of the tracking data structures will be measured and compared against alternatives such as bitmaps.
3. **Concurrency Stress Testing.** The existing concurrent-stream test (`tc06_concurrent_streams`) exercises the xarray under parallel GPU faults. Additional stress tests under multi-process workloads and simultaneous range-filter updates are needed to validate the spinlock boundary.

Optimisation

Based on profiling results, targeted optimisations will be applied. Likely candidates include batching xarray inserts to reduce per-fault locking overhead, and replacing the xarray with a bitmap representation for workloads where the dirty set density is high enough to justify the fixed memory cost.

Further Deliverables

1. **Documentation.** Comprehensive usage documentation covering the procfs interface, epoch semantics, range filtering, and known limitations.
2. **Formal Benchmarking Report.** Quantified results for fault latency overhead, memory footprint, and migration throughput impact across representative workloads, with and without the tracker active.
3. **Command-line Interface.** A user-facing CLI tool that wraps the procfs interface, providing commands to start and stop tracking, query dirty pages for a given PID, set address range filters, and display results in a human-readable format.

Declaration

We declare that the work presented in this report is our own, except where otherwise acknowledged. We have also read and understood the University's policy on plagiarism and academic integrity, and we confirm that this report complies with those standards.